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Auditor-Free Byzantine-Resilient Federated Learning via Functional Inner Products

DEEPAK GUPTA
2024PCS0024

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A Dissertation Submitted to

Indian Institute of Technology, Jammu

In Partial Fulfillment of the Requirements for the Degree of

Master of Technology

in

Computer Science and Engineering



भारतीय प्रौद्योगिकी
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Department of Computer Science & Engineering

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DECLARATION

This thesis is submitted as a requirement for the award of the Master of Technology degree in Computer Science and Engineering in the Department of Computer Science and Engineering. I declare that this written submission represents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the original sources. The research embodied in this thesis has not been submitted in part or full, to any other University or Institute for the award of any other degree or diploma before. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the Institute and can also evoke penal action from the sources that have thus not been properly cited, or from whom proper permission has not been taken when needed.

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Dr. Yamuna Prasad, IIT Jammu
Supervisor

*To my parents,
who gave me the courage to dream big,
and to my teachers,
who showed me how to turn those dreams into reality.*

Acknowledgement

This thesis grew through many runs that failed before they became useful, many small corrections in the protocol, and many discussions that helped me understand the problem more clearly.

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Finally, I thank my parents and my family for their constant support and encouragement throughout this thesis.

Deepak Gupta

List of Publications

Conference Publications

1. Mahendra Kumar Gurve, Deepak Gupta, Nitin, and Yamuna Prasad, “Eliminating the Auditor: Functional Inner Product Based Byzantine-Resilient Federated Learning,” accepted as a poster at the *ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2026)*, Saint-Malo, France, May 2026.

Abstract

2 Federated Learning (FL) lets many clients train a shared model without exposing their raw data. To strengthen the privacy guarantee, modern deployments wrap each client's update in additive homomorphic encryption, giving rise to Privacy-Preserving Federated Learning (PPFL). The same encryption that hides gradients from a curious server, however, also hides whether they are honest: a Byzantine client can flip labels before training and quietly poison the aggregated model, either degrading overall accuracy or installing a backdoor on a chosen target class.

Existing defences in this setting rely on auditor-based architectures that insert a trusted third party to decrypt cluster-level statistics of the incoming updates and screen Byzantine clients with distributional models. While effective, such schemes reintroduce the very trust anchor PPFL was designed to remove, expose coarse structural information about the encrypted gradients, and scale super-linearly in the model dimension. A defence that is simultaneously fully encrypted, free of any new trust anchor, and linear in the model dimension has remained an open challenge.

This thesis develops **FIP-FL**, an auditor-free defence for the encrypted-gradient setting. Instead of clustering decrypted summaries, the verifier computes, under Paillier encryption, one projection score for each client: the *Functional Inner Product* (FIP) between that client's update and a momentum-tracked reference direction. The decrypted scalar then passes through the two checks used in my implementation and poster flow: first a direction filter, then a median-MAD magnitude window. I analyse the resulting protocol in terms of scalar-gradient privacy, the admissible region left to an attacker, and descent under an L -smooth objective.

2 The framework is implemented end-to-end and evaluated on MNIST and KDDCup99, under both IID and non-IID client partitions, and against targeted and untargeted label-flipping attacks at malicious-client rates up to 50%. Across this evaluation grid, FIP-FL achieves strong robustness without any external auditor. The research underlying this thesis has been accepted as a poster at the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2026), Saint-Malo, France, May 2026.

Contents

List of Publications	xi
Abstract	xiii
List of Figures	xvii
List of Tables	xix
List of Abbreviations	xxi
1 Introduction	1
1.1 Background and Context	1
1.2 The Research Gap	1
1.3 Research Objectives and Contributions	2
2 Background and Motivation	3
2.1 Federated Learning and FedAvg	3
2.2 IID and Non-IID Federations	3
2.3 Privacy-Preserving Federated Learning	4
2.4 Poisoning Attack Taxonomy	4
3 Literature Survey	5
3.1 Robust Aggregation Rules	5
3.2 Detection-Based Defences	5
3.3 Auditor-Based PPFL Baseline	6
3.4 Positioning of FIP-FL	6
4 Preliminaries	7
4.1 Notation	7
4.2 The Paillier Cryptosystem	7
4.3 Functional Inner Product	8
4.4 Robust Statistics	9
4.5 System and Threat Model	9
4.6 Security and Privacy Goals	9
5 Proposed Framework: FIP-FL	11
5.1 System Architecture	11
5.2 System Setup and Key Distribution	13

5.3	Per-Round Protocol	15
5.3.1	Phase I: Local Training and Encrypted Update Generation	16
5.3.2	Phase II: Homomorphic FIP Score Computation	18
5.3.3	Phase III: Two-Stage Geometric Audit	20
5.3.4	Phase IV: Secure Aggregation	22
5.3.5	Phase V: Reference Update and Model Distribution	23
5.4	Theoretical Analysis	25
5.4.1	Gradient Privacy	26
5.4.2	Attack Resilience	26
5.4.3	Convergence Guarantee	27
6	Experimental Setup	29
6.1	Datasets	29
6.2	Data Partitioning	30
6.3	Attack Models	30
6.4	Evaluation Metrics	31
6.5	Implementation and Hardware	32
7	Results and Analysis	33
7.1	Non-IID Partition, Untargeted Attacks	33
7.1.1	MNIST	33
7.1.2	KDDCup99	34
7.2	Non-IID Partition, Targeted Attacks	34
7.2.1	MNIST	34
7.2.2	KDDCup99	35
7.3	IID Partition, Untargeted Attacks	36
7.3.1	MNIST	36
7.3.2	KDDCup99	37
7.4	IID Partition, Targeted Attacks	38
7.4.1	MNIST	38
7.4.2	KDDCup99	38
7.5	Aggregate Findings Across the Grid	39
8	Comparative Analysis	41
8.1	Privacy	41
8.2	Computational Complexity	42
8.3	Communication Complexity	42
8.4	Robustness and Engineering Trade-Offs	43
8.5	Scheme-versus-Scheme Accuracy Comparison	43
9	Conclusion and Future Work	45
9.1	Summary of the Thesis	45
9.2	Limitations and Future Work	45

List of Figures

5.1 Geometric architecture of the FIP-FL protocol. 12

7.1 Non-IID untargeted attack dynamics over 300 communication rounds. Accuracy converges stably for both datasets, while the direction/magnitude audit reaches perfect malicious-client detection by the final round. 35

7.2 KDDCup99 non-IID targeted attack results. The target-class trajectory remains high through training, and the round-300 comparison shows that overall and target-class accuracy stay stable even as the malicious-client fraction increases. 36

7.3 IID untargeted attack dynamics over 300 communication rounds. Both datasets show stable convergence, and the audit maintains perfect malicious-client detection without false positives at the reported final round. 37

7.4 IID targeted attack target-class accuracy over 300 communication rounds. In both datasets, the target class recovers to a stable high-accuracy regime by the final round. 39

List of Tables

6.1	Datasets used in the evaluation of FIP-FL.	29
7.1	MNIST, non-IID partition, untargeted label-flipping attack. . .	33
7.2	KDDCup99, non-IID partition (Dirichlet $\alpha = 0.5$), untargeted label-flipping attack.	34
7.3	MNIST, non-IID, targeted label-flipping attack ($c_s = 0, c_t = 7$); SenSys 2026 headline targeted-accuracy cell.	34
7.4	KDDCup99, non-IID partition (2 shards/client), targeted label-flipping attack.	35
7.5	MNIST, IID partition, untargeted label-flipping attack.	36
7.6	KDDCup99, IID partition, untargeted label-flipping attack. . . .	37
7.7	MNIST, IID partition, targeted label-flipping attack ($c_s = 0, c_t = 7$).	38
7.8	KDDCup99, IID partition, targeted label-flipping attack.	38
8.1	Per-round computational cost. N : clients; $ \mathcal{A} $: accepted clients; d : model dimension; B : batch size; E : local epochs.	42
8.2	Per-round communication cost by directed link.	42
8.3	Accuracy on MNIST non-IID under a 50% malicious-client rate. Entries are percentages; baseline values are from [1].	43

List of Abbreviations

ACC — Overall Accuracy

BFV — Brakerski–Fan–Vercauteren (homomorphic encryption scheme)

BGV — Brakerski–Gentry–Vaikuntanathan (homomorphic encryption scheme)

CKKS — Cheon–Kim–Kim–Song (approximate homomorphic encryption)

DCR — Decisional Composite Residuosity

DoS — Denial of Service

DP — Differential Privacy

FedAvg — Federated Averaging

FHE — Fully Homomorphic Encryption

FIP — Functional Inner Product

FIP-FL — Functional Inner Product Federated Learning (proposed framework)

FL — Federated Learning

FPR — False Positive Rate

GMM — Gaussian Mixture Model

HE — Homomorphic Encryption

IID — Independent and Identically Distributed

KDD — Knowledge Discovery in Databases

MAD — Median Absolute Deviation

MNIST — Modified National Institute of Standards and Technology (database)

O_{acc} — Other-label Accuracy

PCA — Principal Component Analysis

PPFL — Privacy-Preserving Federated Learning

R2L — Remote-to-Local (KDDCup attack class)

SenSys — ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems

SGD — Stochastic Gradient Descent

SHA — Secure Hash Algorithm

SMPC — Secure Multiparty Computation

T_{acc} — Target-class Accuracy

TF-IDF — Term Frequency – Inverse Document Frequency

TKA — Trusted Key Authority

TPR — True Positive Rate

U2R — User-to-Root (KDDCup attack class)

1 Introduction

1.1 Background and Context

Federated Learning (FL), introduced by McMahan et al. [2], is useful when data owners can train a shared model but cannot pool their records. In the setting studied here, clients keep MNIST or KDDCup99 samples locally, compute a gradient update, and send only that update to the server.

The update itself is still sensitive. Gradients can leak training examples [3], membership information, or distributional attributes, so Privacy-Preserving Federated Learning (PPFL) protects the update channel through differential privacy, secure aggregation, or homomorphic encryption. This thesis uses additive homomorphic encryption: the server can add encrypted updates, but it cannot inspect any client's gradient coordinates.

1.2 The Research Gap

The same encryption that protects the update also removes the usual evidence used for Byzantine defence. A malicious client can train on flipped labels or send a scaled poison update while the server receives only ciphertexts. Krum [8], coordinate-wise median and trimmed-mean [9], and Bulyan [10] need plaintext distances or coordinate order statistics; Auror [11], FL-Trust [12], and PEFL [13] also depend on plaintext updates or revealed summaries.

The closest encrypted-gradient baseline for this thesis is the auditor-based scheme of Yazdinejad et al. [1]. It performs well, but it introduces a trusted auditor, decrypts cluster-level summaries, pays an $\mathcal{O}(d^3)$ covariance-inversion cost, and can reveal population structure when client data are non-IID. The gap

I address is therefore narrow and concrete: Byzantine filtering over encrypted individual updates, without an auditor, with verifier work close to linear in the model dimension.

1.3 Research Objectives and Contributions

This thesis designs, analyses, and evaluates FIP-FL, an auditor-free PPFL framework for label-flipping attacks. FIP-FL verifies each encrypted update by computing a functional inner product (FIP) with a trusted reference direction. Only this scalar score is decrypted, and the score is checked by a two-stage geometric audit: first direction, then magnitude.

The main contributions are:

1. **Auditor-free encrypted verification.** FIP-FL replaces decrypted cluster statistics with one homomorphic projection score per client.
2. **Two-stage geometric audit.** A direction filter removes label-flip style reversals, and a median–MAD magnitude window removes weak or over-scaled positive scores before aggregation.
3. **Formal analysis.** The thesis proves what is and is not revealed by the scalar score, characterises the accepted attack region, states a descent condition under L -smoothness, and compares verifier complexity with the auditor-based baseline.
4. **Empirical validation.** The experiments cover MNIST and KDDCup99, IID and non-IID partitions, targeted and untargeted label flipping, and malicious-client rates up to 50%.
5. **Engineering efficiency.** The verifier uses $\mathcal{O}(Nd + N \log N)$ work instead of an $\mathcal{O}(d^3)$ covariance step, and the parallel Phase II implementation gives a 13× speed-up on the 25-core workstation.

The research underlying this thesis has been accepted as a poster at the ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys 2026), Saint-Malo, France, May 2026.

2

2 Background and Motivation

2.1 Federated Learning and FedAvg

2

Consider N clients, where client i holds private data $\mathcal{D}_i = \{(x_k^{(i)}, y_k^{(i)})\}_{k=1}^{n_i}$ and $n = \sum_i n_i$. The federated objective is the weighted empirical risk

$$F(\mathbf{w}) = \sum_{i=1}^N \frac{n_i}{n} F_i(\mathbf{w}), \quad F_i(\mathbf{w}) = \frac{1}{n_i} \sum_{k=1}^{n_i} \ell(f_{\mathbf{w}}(x_k^{(i)}), y_k^{(i)}). \quad (2.1)$$

FedAvg [2] optimises this objective by broadcasting $\mathbf{w}^{(t)}$, running local SGD at each client, and aggregating returned updates. With descent-direction update $\Delta\theta_i^{(t)} = \mathbf{w}^{(t)} - \mathbf{w}_i^{(t+1)}$, the server update is

11

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \sum_{i=1}^N \frac{n_i}{n} \Delta\theta_i^{(t)}. \quad (2.2)$$

The server therefore needs only coordinate-wise aggregation, while raw samples remain local.

2.2 IID and Non-IID Federations

FedAvg is simplest when client data are IID draws from a common population. Real federations are usually non-IID: users, hospitals, and network sites observe different label distributions and feature patterns. This thesis uses label skew, where each client receives a subset of labels with controlled overlap. Label skew makes local objectives F_i drift from the global objective F , so honest gradients can differ substantially. A Byzantine defence must therefore reject malicious deviation without treating legitimate heterogeneity as an attack.

2.3 Privacy-Preserving Federated Learning

Gradient privacy is not automatic in FL. Reconstruction, membership inference, and property inference attacks can exploit the update vectors themselves [3]. PPFL therefore adds a protection layer to the update channel. Differential privacy clips and noises the update, which can hurt utility. Secure aggregation [14] hides individual updates from the server, but normally exposes only the final sum, after per-client screening would have been needed. Homomorphic encryption permits limited computation on ciphertexts: Paillier [4] gives encrypted addition and multiplication by known plaintext scalars, whereas BFV and CKKS [5] support richer packed arithmetic with a heavier implementation cost.

FIP-FL uses Paillier because the verifier needs only a linear operation. An encrypted client update is projected onto a plaintext reference direction, giving one homomorphic inner product score. This choice keeps the audit compatible with encrypted updates without adding DP noise or requiring a peer-to-peer secure-aggregation round.

2.4 Poisoning Attack Taxonomy

A Byzantine client can submit any valid-looking ciphertext vector, so the server cannot rely on syntax checks. This thesis evaluates two pre-training label-flipping attacks. In the *untargeted* case, the malicious client remaps every label by a fixed cyclic offset and tries to reduce overall accuracy. In the *targeted* case, it changes one source class to one target class, so the global accuracy can remain plausible while a chosen class fails. Sign flipping, scaling, model replacement, and collusion are not used as separate experiment rows here, but they motivate the two audit components: a sign/direction check and a magnitude window.

3 Literature Survey

3.1 Robust Aggregation Rules

2 Robust aggregation changes the FedAvg mean into a statistic that is less sensitive to adversarial updates. Krum and Multi-Krum [8] choose updates that are close to many neighbours, which requires plaintext pairwise distances and costs $\mathcal{O}(N^2d)$. Coordinate-wise median and trimmed-mean [9] reduce the cost to $\mathcal{O}(Nd \log N)$, but label skew can make honest coordinates multi-modal rather than tightly centered. Bulyan [10] combines Krum-style selection with trimmed averaging, improving robustness while still requiring plaintext access and sufficient honest-client margins.

These methods are useful reference points, but they are not drop-in PPFL solutions. Over encrypted updates, the server cannot sort coordinates, compute distances, or select nearest neighbours unless some party decrypts additional information.

3.2 Detection-Based Defences

Detection-based defences filter clients before aggregation. Auror [11] clusters update features and rejects minority clusters; under non-IID data, honest clients can also split into several clusters. FL-Trust [12] uses a trusted server gradient to score client updates, which is close in spirit to FIP-FL, but it assumes plaintext updates. PEFL [13] brings correlation scoring into an encrypted setting, although the auditor-based study reports weaker robustness at high attack rates and under non-IID partitions [1]. FoolsGold [15] relies on historical similarity among colluding clients, and ShieldFL [16] uses helper-assisted secure

computation.

For this thesis, the important trade-off is visible across these methods: stronger detection often brings plaintext access, another trusted party, or additional protocol communication. FIP-FL is positioned at the narrower point where individual updates remain encrypted and the verifier receives only one score per client.

3.3 Auditor-Based PPFL Baseline

Yazdinejad et al. [1] are the main baseline because they study Byzantine-resilient FL in an encrypted-update pipeline. Their design inserts a trusted auditor between clients and the aggregation server. The auditor decrypts cluster-level summaries, fits a Gaussian mixture model after projection, and applies Mahalanobis-distance screening before trusted updates are forwarded for homomorphic aggregation.

This baseline is strong empirically, reporting about 96.5% target-class accuracy on MNIST non-IID at a 50% targeted label-flipping rate. The limitations that motivate FIP-FL are structural: an extra trust anchor is introduced, covariance inversion costs $\mathcal{O}(d^3)$, and decrypted cluster statistics can expose non-IID population structure.

3.4 Positioning of FIP-FL

FIP-FL differs from the surveyed work in four concrete ways. It keeps individual updates encrypted, introduces no external auditor or helper, replaces distributional clustering with alignment to a trusted reference direction, and keeps verifier work linear in d apart from sorting the scalar scores. Compared with plaintext robust aggregation it preserves encrypted updates; compared with auditor-based PPFL it removes decrypted cluster statistics and the cubic Mahalanobis step; compared with root-gradient methods it computes the alignment score homomorphically. The next chapter fixes the notation and primitives used by this design.

4 Preliminaries

4.1 Notation

Scalars are lowercase italics (a, η), vectors are lowercase bold ($\mathbf{x}, \boldsymbol{\theta}$), and matrices are uppercase bold ($\mathbf{W}, \mathbf{M}$). For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$,

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{k=1}^d a_k b_k, \quad \|\mathbf{a}\|_2 = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}. \quad (4.1)$$

A Paillier ciphertext is written $\llbracket m \rrbracket = \text{Enc}_{pk}(m)$. The main FL symbols are N clients, model dimension d , round index t , local epochs E , batch size B , learning rate η , and Paillier security parameter λ .

4.2 The Paillier Cryptosystem

Paillier [4] is the additive homomorphic scheme used in the implementation. For security parameter λ , the Trusted Key Authority samples primes p, q , sets $n = pq$, uses generator $g = n + 1$, and keeps the secret key $sk = (\ell, \alpha)$ with $\ell = \text{lcm}(p-1, q-1)$ and $\alpha = \ell^{-1} \bmod n$. In all experiments, $\lambda = 1024$.

For plaintext $m \in \mathbb{Z}_n$ and random $r \in \mathbb{Z}_n^*$,

$$\text{Enc}_{pk}(m) = g^m r^n \pmod{n^2}, \quad (4.2)$$

and decryption is

$$\text{Dec}_{sk}(c) = L(c^\ell \bmod n^2) \alpha \pmod{n}, \quad L(x) = \frac{x-1}{n}. \quad (4.3)$$

For plaintexts m_1, m_2 and known integer a ,

$$\text{Enc}_{pk}(m_1)\text{Enc}_{pk}(m_2) \equiv \text{Enc}_{pk}(m_1 + m_2) \pmod{n^2}, \quad (4.4)$$

$$\text{Enc}_{pk}(m_1)^a \equiv \text{Enc}_{pk}(am_1) \pmod{n^2}. \quad (4.5)$$

These two identities are exactly the operations required by FIP-FL: summing encrypted update coordinates and multiplying an encrypted coordinate by a known reference-coordinate scalar. No ciphertext–ciphertext multiplication is required.

Real gradients are encoded as fixed-point integers. For scale s ,

$$\text{encode}(x) = \text{round}(sx), \quad \text{decode}(z) = z/s. \quad (4.6)$$

Signed values use centered modular encoding in the Paillier plaintext ring: $-a$ is stored as $n - a$ and decoded from $(-n/2, n/2]$. When a plaintext scalar in homomorphic multiplication is negative, the implementation uses the corresponding modular inverse. The fixed-point scale is $s = 10,000$. Because a FIP score multiplies an encoded gradient coordinate by an encoded reference coordinate, the scalar decrypted by the TKA is decoded by dividing by s^2 .

4.3 Functional Inner Product

For gradient vectors, the Functional Inner Product used in this thesis is the Euclidean dot product

$$\langle \mathbf{g}_i, \mathbf{g}_{\text{ref}} \rangle = \sum_{k=1}^d g_i^{(k)} g_{\text{ref}}^{(k)}. \quad (4.7)$$

It measures whether a client update aligns with the trusted reference direction. Combining (4.7) with Paillier scalar multiplication and addition gives the homomorphic score

$$\llbracket \langle \mathbf{g}_i, \mathbf{g}_{\text{ref}} \rangle \rrbracket = \prod_{k=1}^d \llbracket g_i^{(k)} \rrbracket^{\text{encode}(g_{\text{ref}}^{(k)})} \pmod{n^2}. \quad (4.8)$$

The TKA decrypts only this scalar. The server never observes a coordinate of $\mathbf{g}_i$.

4.4 Robust Statistics

The verifier thresholds one scalar score per client; therefore the robust statistics are one-dimensional. For score set $\mathcal{S} = \{s_1, \dots, s_N\}$,

$$\tilde{\mu} = \text{median}(\mathcal{S}), \quad \text{MAD} = \text{median}\{|s_i - \tilde{\mu}| : s_i \in \mathcal{S}\}. \quad (4.9)$$

The median and MAD are used because they remain stable until contamination approaches the 50% boundary. FIP-FL uses the adaptive window

$$\tau_{\min}^{(t)} = \alpha(\rho^{(t)})\tilde{\mu}^{(t)}, \quad \tau_{\max}^{(t)} = \tilde{\mu}^{(t)} + \gamma_{\text{mad}}\text{MAD}^{(t)}, \quad (4.10)$$

with $\gamma_{\text{mad}} = 5$. A score below the lower floor contributes too weakly along the reference direction, whereas a score above the upper cap is aligned but carries unusually large influence.

4.5 System and Threat Model

FIP-FL uses four roles. Clients train locally and submit encrypted gradients. The Aggregation Server combines only accepted ciphertexts and broadcasts the model. The FIP Verifier computes scalar FIP scores and applies the two-stage audit. The Trusted Key Authority generates the Paillier keys and decrypts only the protocol-specified scalar scores or aggregate updates. The Verifier can be deployed with the server, but it does not become an external auditor because it never receives plaintext client gradients.

The adversary controls up to $f < N/2$ Byzantine clients. It may relabel local data, submit arbitrary valid ciphertexts, collude across malicious clients, and observe public broadcasts. The experiments instantiate this model with targeted and untargeted label flipping at malicious rates $f/N \in \{10\%, 20\%, 30\%, 50\%\}$.

4.6 Security and Privacy Goals

The protocol is designed around four goals. *Privacy*: the server and Verifier learn no client-gradient coordinate, norm, or pairwise distance beyond the scalar score

$s_i^{(t)} = \langle \mathbf{g}_i^{(t)}, \mathbf{g}_{\text{ref}}^{(t)} \rangle$. *Byzantine resilience*: accepted updates must have a positive reference projection and a score inside $(\tau_{\min}^{(t)}, \tau_{\max}^{(t)})$. *Convergence*: the accepted aggregate should satisfy the descent alignment and step-size condition stated in Chapter 5. *Efficiency*: verifier work should remain $\mathcal{O}(Nd + N \log N)$, and aggregation should cost $\mathcal{O}(|\mathcal{A}^{(t)}|d)$ for accepted set $\mathcal{A}^{(t)}$.

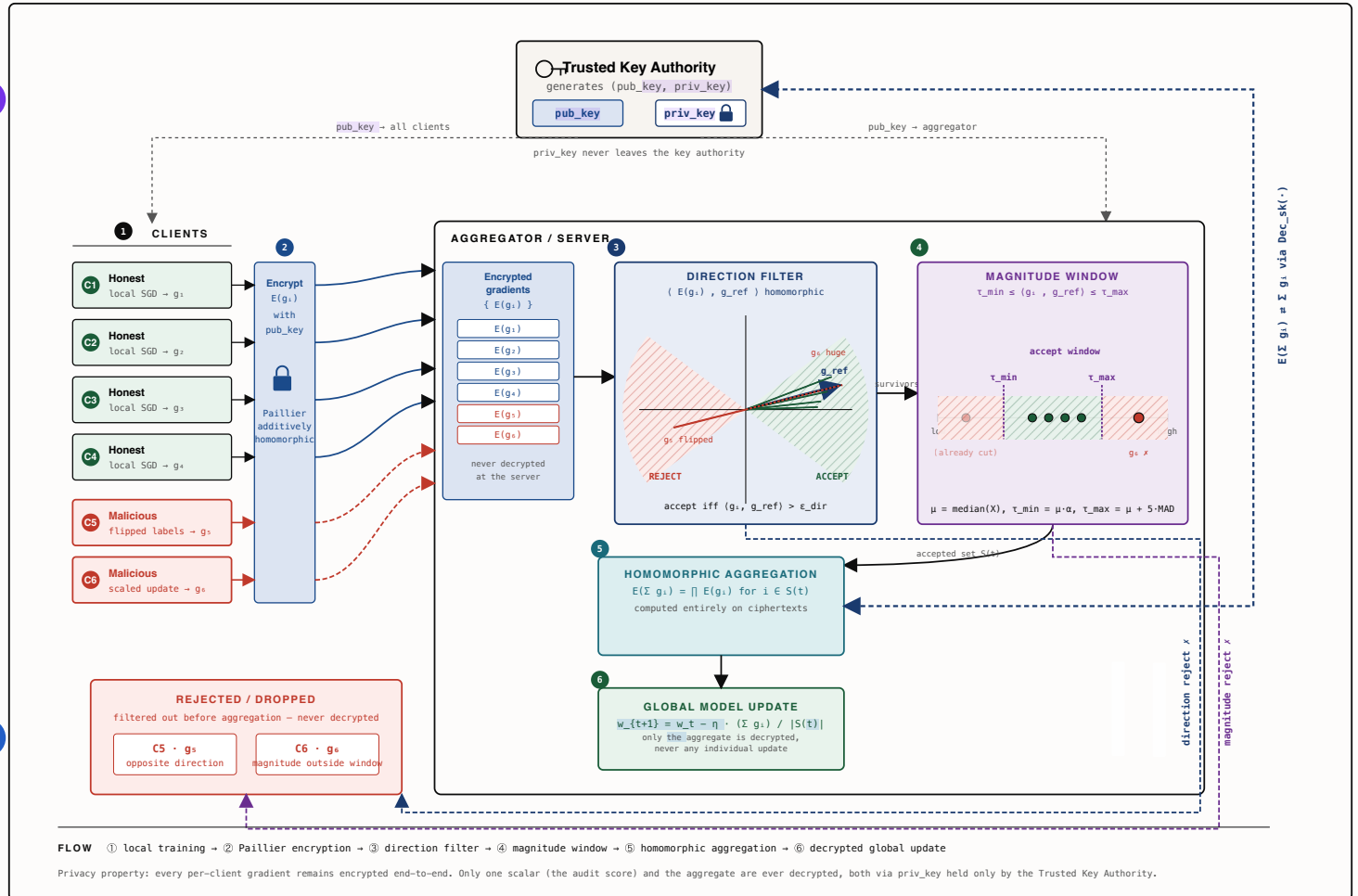
5 Proposed Framework: FIP-FL

5.1 System Architecture

FIP-FL has four logical entities: N clients, an Aggregation Server, a FIP Verifier, and a Trusted Key Authority (TKA). Figure 5.1 shows the round pipeline: encrypted client updates, one FIP score per client, filtering, accepted-ciphertext aggregation, and TKA decryption only for scalar scores or the final aggregate.

FIP-PPFL — Functional Inner Product · Privacy-Preserving Federated Learning

A trusted key authority publishes a Paillier public key. Clients train locally and submit encrypted gradients. The aggregator audits each encrypted update against a benign reference using two geometric criteria — first a directional check, then a magnitude window — and homomorphically aggregates only the survivors. The single private key never leaves the key authority.



① TRUSTED KEY AUTHORITY

Generates one Paillier keypair. Distributes the public key to all clients and the aggregator; keeps the private key sealed and uses it only for decryption of the audit score and the aggregate.

$pub_key \rightarrow$ broadcast
 $priv_key \rightarrow$ never leaves

③ DIRECTION FILTER

Server computes the homomorphic inner product $\langle g_i, g_ref \rangle$ against a benign reference built on public data. Updates pointing in the wrong half-space are removed before any aggregation.

accept iff $\langle g_i, g_ref \rangle > \epsilon_dir$

④ MAGNITUDE WINDOW

Robust statistics on the alignment scores. Median μ and MAD define an adaptive window; any client whose score is too small or too large is dropped — symmetric protection against shrinking and scaling attacks.

$\tau_min \leq \langle g_i, g_ref \rangle \leq \tau_max$

⑤ SECURE AGGREGATION

The encrypted gradients of the surviving set $S(t)$ are summed homomorphically. Only the aggregate is sent to the key authority for decryption — individual gradients never appear in plaintext.

$E(\sum g_i) = \sum E(g_i)$, $i \in S(t)$

FIP-PPFL · System flow · Paillier additively homomorphic encryption

Geometric audit · direction + magnitude · no plaintext leak

This split keeps the privacy boundary explicit: the Verifier sees encrypted updates and the plaintext reference $\mathbf{R}^{(t)}$, but no plaintext client gradient; the TKA decrypts scalar FIP scores and aggregate updates, but does not run the audit.

5.2 System Setup and Key Distribution

Setup runs once before round 1. Algorithm 1 separates Paillier state, model state, and geometric state: the TKA creates the encryption domain, the server initializes the model, and the Verifier stores a unit-norm reference from public bootstrap data.

FIP-PPFL System Initialization and Reference Bootstrapping

A geometric view of the three setup phases — key generation & registration, model initialization, and reference bootstrapping — culminating in a unit-norm trusted reference direction $g_{\text{ref}}^{(o)} \in \mathbb{R}^d$ stored at the server-side FIP Verifier module for round 1.

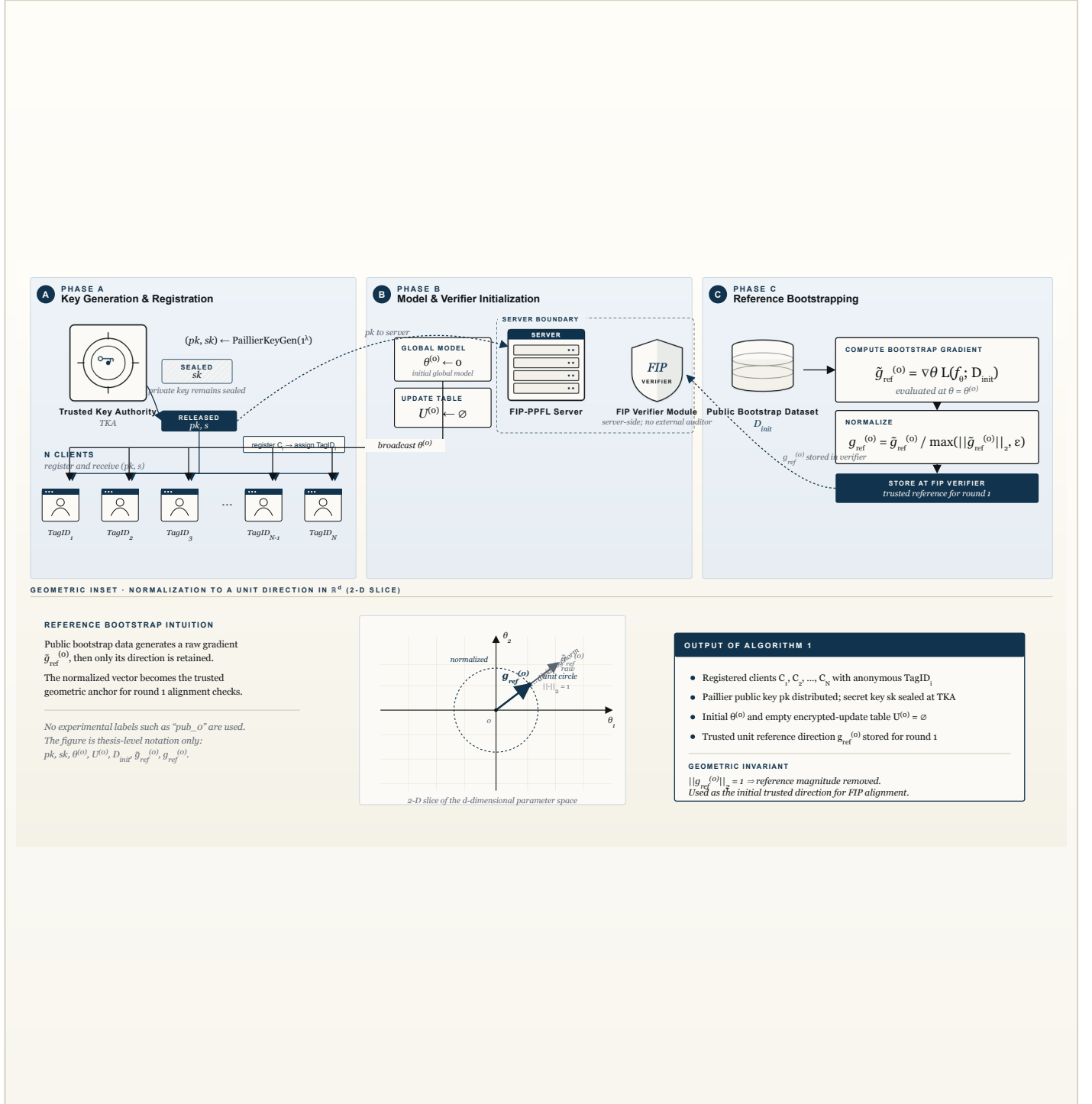


Figure 1. Algorithm 1. The TKA generates the Paillier key pair and keeps sk sealed while releasing only pk and the fixed-point scale S . The server initializes $\theta^{(o)} \leftarrow \text{InitModel}(f_0)$ (zero initialization for the logistic-regression experiments) together with $U^{(o)} = \emptyset$, broadcasts $\theta^{(o)}$ to the registered clients, and the server-side FIP Verifier stores the initial trusted reference direction. The trusted public bootstrap set D_{init} produces the raw gradient $\tilde{g}_{\text{ref}}^{(o)}$, which is normalized to $g_{\text{ref}}^{(o)}$ using $\max(\|\tilde{g}_{\text{ref}}^{(o)}\|_2, \epsilon)$. The bottom inset shows the geometric normalization from a raw gradient vector to a unit direction in a 2-D slice of the parameter

Algorithm 1 FIP-PPFL System Initialization and Reference Bootstrapping

Require: Number of clients N , model f_{θ} , public bootstrap set $\mathcal{D}_{\text{init}}$, security parameter λ , fixed-point scale s

Ensure: Registered clients, Paillier keys, initial global model $\theta^{(0)}$, trusted reference direction $\mathbf{g}_{\text{ref}}^{(0)}$

Key generation and registration

- 1: $(pk, sk) \leftarrow \text{PAILLIERKEYGEN}(1^\lambda)$ at the Trusted Key Authority
- 2: TKA keeps sk sealed and releases only pk to the clients and the FIP-PPFL server

3: **for** $i \leftarrow 1$ **to** N **do**

4: Register client C_i and assign anonymous tag TagID_i

5: Send (pk, s) to client C_i

6: **end for**

Model and verifier initialisation

- 7: Initialise global model parameters $\theta^{(0)} \leftarrow \mathbf{0}$
- 8: Broadcast $\theta^{(0)}$ to all registered clients
- 9: Initialise encrypted-update table $\mathcal{U}^{(0)} \leftarrow \emptyset$

Reference bootstrapping

- 10: Compute the trusted bootstrap gradient on public data:
- 11: $\tilde{\mathbf{g}}_{\text{ref}}^{(0)} \leftarrow \nabla_{\theta} \mathcal{L}(f_{\theta}; \mathcal{D}_{\text{init}}) \big|_{\theta=\theta^{(0)}}$
- 12: Normalise the reference direction:
- 13: $\mathbf{g}_{\text{ref}}^{(0)} \leftarrow \tilde{\mathbf{g}}_{\text{ref}}^{(0)} / \max\left(\left\|\tilde{\mathbf{g}}_{\text{ref}}^{(0)}\right\|_2, \varepsilon\right)$
- 14: Store $\mathbf{g}_{\text{ref}}^{(0)}$ at the FIP Verifier for round 1

The bootstrap set is public and non-sensitive; it only provides $\mathbf{g}_{\text{ref}}^{(0)}$ at the zero model. From round t onward, the same object is denoted $\mathbf{R}^{(t)}$.

5.3 Per-Round Protocol

Each round $t \geq 1$ follows the poster flow: encrypted gradient, FIP score, direction filter, magnitude window, secure aggregation, and reference update.

5.3.1 Phase I: Local Training and Encrypted Update Generation

Each client trains for E local epochs, averages mini-batch gradients, flattens weights and bias into a d -dimensional vector, encrypts each coordinate, and sends the ciphertext vector to the Verifier. Algorithm 2 describes this path; malicious clients first alter local labels.

Client-Side Local Training and Encrypted Gradient Submission

A geometric view of the client-side round procedure — receiving the round model, performing local mini-batch training, averaging and flattening the local gradient, encoding each coordinate with fixed-point scale S , encrypting under Paillier public key pk , and sending the anonymous encrypted submission $M_i^{(t)}$ to the FIP Verifier.

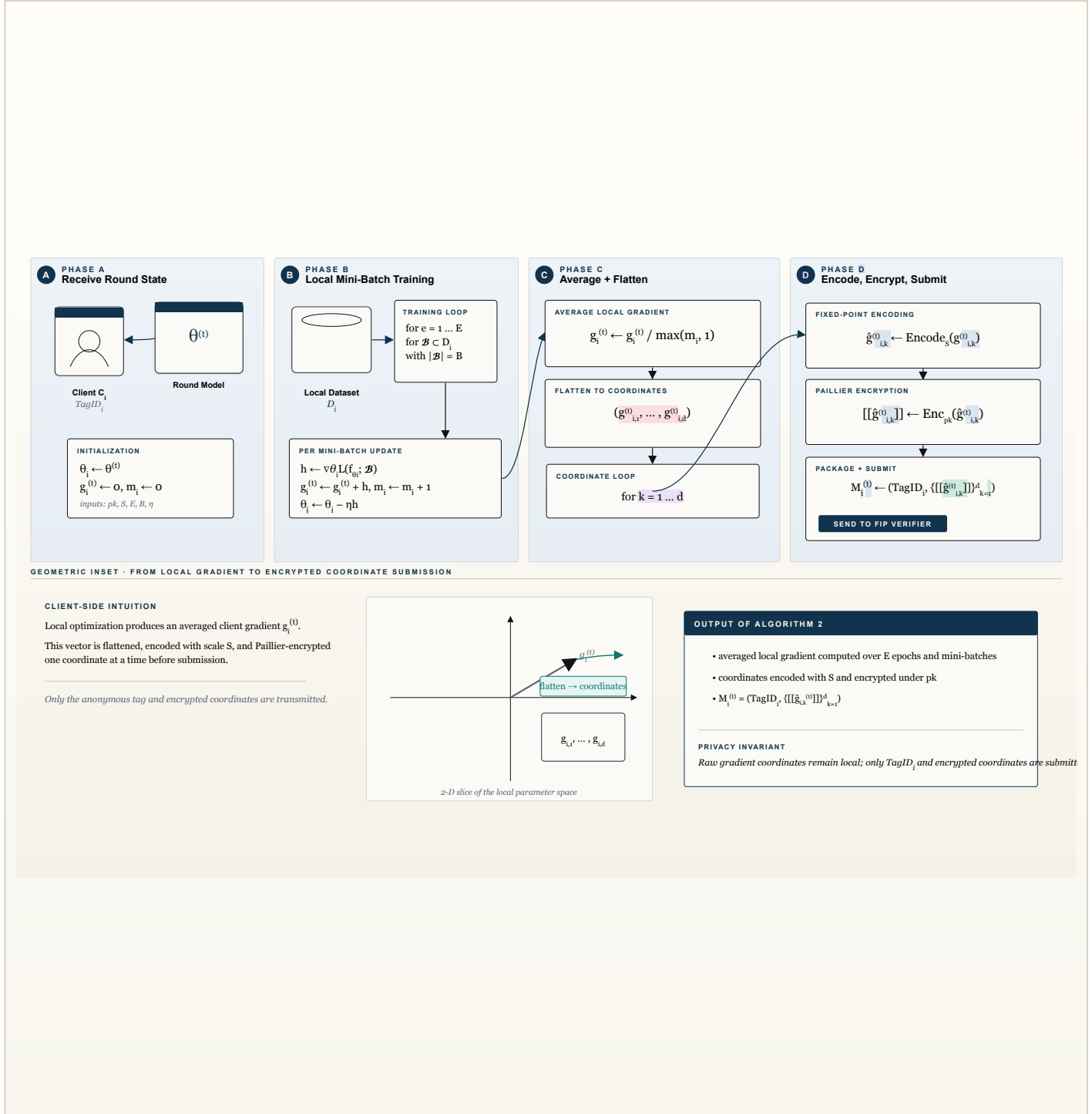


Figure 2. Algorithm 2. Client C_i receives the round model $\theta^{(t)}$, public key pk , fixed-point scale S , and local hyperparameters. It initializes $\theta_i \leftarrow \theta^{(t)}$, performs local mini-batch SGD over E epochs, accumulates and averages the local gradient $g_i^{(t)}$, and then flattens it into coordinates. Each coordinate is encoded as $\hat{g}_{i,k}^{(t)}$ using S and encrypted as $[[\hat{g}_{i,k}^{(t)}]]$ under pk . The transmitted message $M_i^{(t)} = (\text{TagID}_i, \{[[\hat{g}_{i,k}^{(t)}]]\}_{k=1}^d)$ contains only the anonymous tag and encrypted encoded coordinates. The inset summarizes the local descent direction and the flatten-encode-encrypt pipeline.

Algorithm 2 Client-Side Local Training and Encrypted Gradient Submission

Require: Client C_i with data $\mathcal{D}_i$, round model $\theta^{(t)}$, public key pk , scale s , epochs E , batch size B , learning rate η

Ensure: Encrypted submission $\mathcal{M}_i^{(t)}$

```

1:  $\theta_i \leftarrow \theta^{(t)}$ 
2:  $\mathbf{g}_i^{(t)} \leftarrow \mathbf{0}$ ,  $m_i \leftarrow 0$ 
3: for  $e \leftarrow 1$  to  $E$  do
4:   for all mini-batches  $\mathcal{B} \subset \mathcal{D}_i$  of size  $B$  do
5:      $\mathbf{h} \leftarrow \nabla_{\theta} \mathcal{L}(f_{\theta_i}; \mathcal{B})$ 
6:      $\mathbf{g}_i^{(t)} \leftarrow \mathbf{g}_i^{(t)} + \mathbf{h}$ ,  $m_i \leftarrow m_i + 1$ 
7:      $\theta_i \leftarrow \theta_i - \eta \mathbf{h}$ 
8:   end for
9: end for
10:  $\mathbf{g}_i^{(t)} \leftarrow \mathbf{g}_i^{(t)} / \max(m_i, 1)$ 
11: Flatten  $\mathbf{g}_i^{(t)}$  into coordinates  $(g_{i,1}^{(t)}, \dots, g_{i,d}^{(t)})$ 
12: for  $k \leftarrow 1$  to  $d$  do
13:    $\hat{g}_{i,k}^{(t)} \leftarrow \text{ENCODE}_s(g_{i,k}^{(t)})$ 
14:    $\llbracket g_{i,k}^{(t)} \rrbracket \leftarrow \text{ENC}_{pk}(\hat{g}_{i,k}^{(t)})$ 
15: end for
16:  $\mathcal{M}_i^{(t)} \leftarrow (\text{TagID}_i, \{\llbracket g_{i,k}^{(t)} \rrbracket\}_{k=1}^d)$ 
17: Send  $\mathcal{M}_i^{(t)}$  to the FIP Verifier

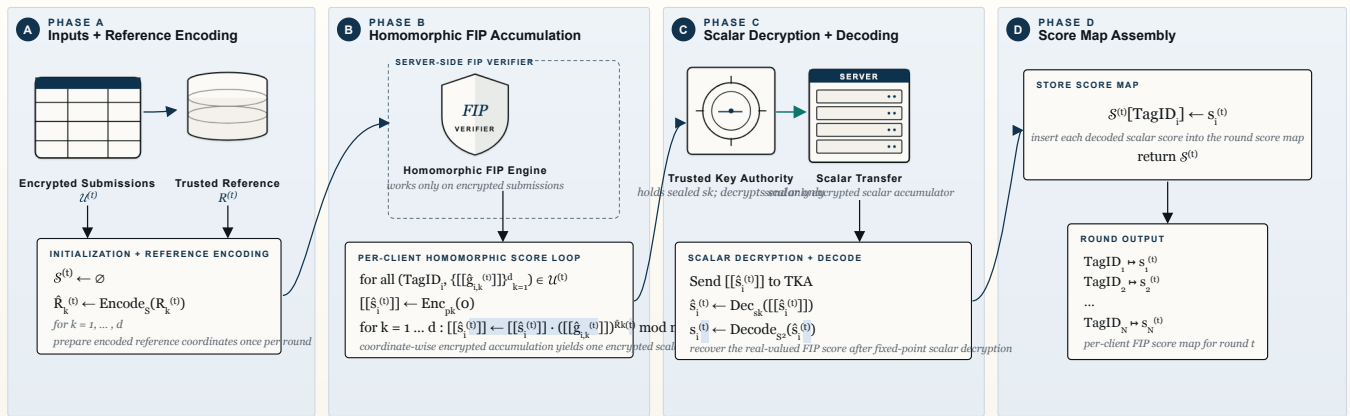
```

5.3.2 Phase II: Homomorphic FIP Score Computation

The Verifier projects each encrypted update onto plaintext reference $\mathbf{R}^{(t)}$ using Paillier's additive homomorphism. The TKA decrypts only $s_i^{(t)} = \langle \mathbf{g}_i^{(t)}, \mathbf{R}^{(t)} \rangle$; no coordinate of $\mathbf{g}_i^{(t)}$ is decrypted. Algorithm 3 gives the computation.

Homomorphic Functional Inner Product Score Computation

A geometric view of server-side encrypted scoring — the FIP Verifier combines encrypted encoded client coordinates with the encoded trusted reference, decrypts only the scalar accumulator through the TKA, decodes the signed fixed-point scalar score, and returns the round score map $\mathcal{S}^{(t)}$.



GEOMETRIC INSET · ENCRYPTED ALIGNMENT WITH A TRUSTED REFERENCE

HOMOMORPHIC FIP INTUITION

For each client, the verifier combines encrypted encoded coordinates with the encoded trusted reference to obtain one encrypted scalar that represents the functional inner product alignment score.

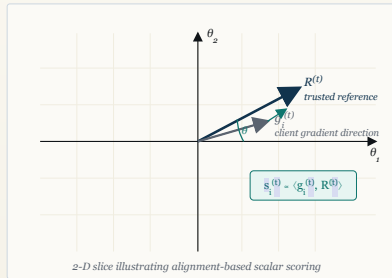
Only this scalar accumulator is decrypted by the TKA; full vectors remain encrypted throughout the server-side computation.

This figure uses thesis-level notation only:

$\mathcal{U}^{(t)}, R^{(t)}, pk, sk, S, \hat{R}_k^{(t)}, [[\hat{g}_{ik}^{(t)}]], [[\hat{s}_i^{(t)}]], s_i^{(t)}, \mathcal{S}^{(t)}$.

SIGNED EXPONENT HANDLING

For negative encoded reference values $\hat{R}_k^{(t)} < 0$, Paillier exponentiation uses ciphertext inversion: $c^a = (c^{-1})^{|a|} \bmod n^2$.



OUTPUT OF ALGORITHM 3

- Reference coordinates $\hat{R}_k^{(t)}$ encoded once for the round
- One encrypted scalar accumulator $[[\hat{s}_i^{(t)}]]$ computed for each TagID_i
- TKA decrypts only the scalar accumulator, not the full encrypted gradient vector
- Decoded score inserted as $\mathcal{S}^{(t)}[\text{TagID}_i] = s_i^{(t)}$
- Return the complete round score map $\mathcal{S}^{(t)}$

PRIVACY INVARIANT

Client vectors remain encrypted end-to-end; only the homomorphic scalar FIP score is decrypted.

Figure 3. Algorithm 3. The FIP Verifier receives the encrypted submission set $\mathcal{U}^{(t)}$ and trusted reference $R^{(t)}$. It initializes the score map $\mathcal{S}^{(t)} = \emptyset$, encodes the reference coordinates as $\hat{R}_k^{(t)}$, and for each client accumulates an encrypted scalar inner-product score $[[\hat{s}_i^{(t)}]]$ using Paillier homomorphism. Only this scalar accumulator is sent to the TKA for decryption; the TKA returns $\hat{s}_i^{(t)}$, which is decoded as $s_i^{(t)}$ using $\text{Decode}_{S^2}(\cdot)$. The verifier stores the result as $\mathcal{S}^{(t)}[\text{TagID}_i] \leftarrow s_i^{(t)}$. The inset illustrates alignment-based scalar scoring with respect to the

Algorithm 3 Homomorphic Functional Inner Product Score Computation

Require: Encrypted submissions $\mathcal{U}^{(t)}$, reference $\mathbf{R}^{(t)}$, public key pk , secret key sk held by TKA, scale s

Ensure: FIP score map $\mathcal{S}^{(t)}$

```

1:  $\mathcal{S}^{(t)} \leftarrow \emptyset$ 
2: Encode reference coordinates  $\widehat{R}_k^{(t)} \leftarrow \text{ENCODE}_s(R_k^{(t)})$  for  $k = 1, \dots, d$ 
3: for all  $(\text{TagID}_i, \{\llbracket g_{i,k}^{(t)} \rrbracket\}_{k=1}^d) \in \mathcal{U}^{(t)}$  do
4:    $\llbracket \widehat{s}_i^{(t)} \rrbracket \leftarrow \text{ENC}_{pk}(0)$ 
5:   for  $k \leftarrow 1$  to  $d$  do
6:      $\llbracket \widehat{s}_i^{(t)} \rrbracket \leftarrow \llbracket \widehat{s}_i^{(t)} \rrbracket \cdot (\llbracket g_{i,k}^{(t)} \rrbracket)^{\widehat{R}_k^{(t)}} \bmod n^2$ 
7:   end for
8:   Send  $\llbracket \widehat{s}_i^{(t)} \rrbracket$  to TKA for scalar decryption
9:    $\widehat{s}_i^{(t)} \leftarrow \text{DEC}_{sk}(\llbracket \widehat{s}_i^{(t)} \rrbracket)$ 
10:   $s_i^{(t)} \leftarrow \text{DECODE}_{s^2}(\widehat{s}_i^{(t)})$ 
11:   $\mathcal{S}^{(t)}[\text{TagID}_i] \leftarrow s_i^{(t)}$ 
12: end for
13: return  $\mathcal{S}^{(t)}$ 

```

5.3.3 Phase III: Two-Stage Geometric Audit

The audit uses two checks. The *direction filter* rejects non-positive FIP scores. Among surviving positive scores, the *magnitude window* accepts only $\tau_{\min}^{(t)} \leq s_i^{(t)} \leq \tau_{\max}^{(t)}$. The lower floor removes weak positive updates, and the upper cap removes unusually large aligned updates. Algorithm 4 gives the procedure.

Two-Stage Geometric Audit for Malicious Client Detection

A geometric view of the audit logic — Stage 1 rejects directionally weak or negatively aligned clients using ϵ_{dir} , while Stage 2 applies robust median/MAD magnitude screening to the provisional pass set and returns accepted encrypted submissions $\mathcal{A}^{(t)}$ and rejected tags $\mathcal{D}^{(t)}$.

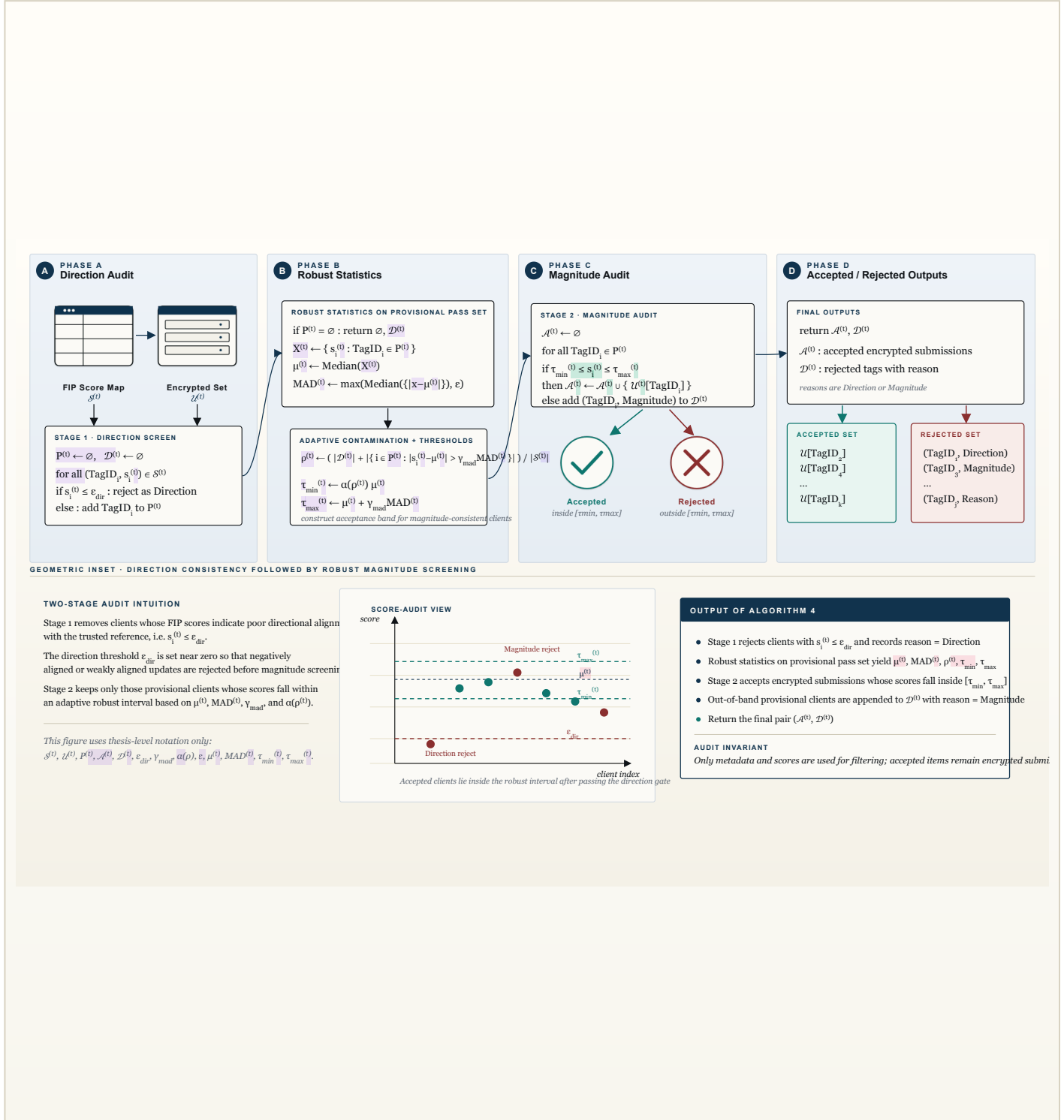


Figure 4. Algorithm 4. The verifier takes the FIP score map $\mathcal{S}^{(t)}$ and encrypted submission set $\mathcal{U}^{(t)}$. In Stage 1, it rejects clients with $s_i^{(t)} \leq \epsilon_{\text{dir}}$ as directionally suspicious and forms the provisional pass set $P^{(t)}$. In Stage 2, it computes the robust statistics $\mu^{(t)}$ and $\text{MAD}^{(t)}$, estimates the contamination rate $\rho^{(t)}$, and constructs the adaptive interval $[\tau_{\min}^{(t)}, \tau_{\max}^{(t)}]$. Encrypted submissions whose scores fall inside this band are retained in the accepted set $\mathcal{A}^{(t)}$, while the others are added to the rejected tag set $\mathcal{D}^{(t)}$ with reason Direction or Magnitude. The inset visualizes this two-stage screening in score space.

Algorithm 4 Two-Stage Geometric Audit for Malicious Client Detection**Require:** FIP scores $\mathcal{S}^{(t)}$, encrypted submissions $\mathcal{U}^{(t)}$, direction threshold ϵ_{dir} ,MAD multiplier γ_{mad} , floor map $\alpha(\rho)$, stability floor ε **Ensure:** Accepted encrypted set $\mathcal{A}^{(t)}$, rejected tag set $\mathcal{D}^{(t)}$

```

1:  $\mathcal{P}^{(t)} \leftarrow \emptyset, \mathcal{D}^{(t)} \leftarrow \emptyset$ 
2: for all  $(\text{TagID}_i, s_i^{(t)}) \in \mathcal{S}^{(t)}$  do
3:   if  $s_i^{(t)} \leq \epsilon_{\text{dir}}$  then
4:      $\mathcal{D}^{(t)} \leftarrow \mathcal{D}^{(t)} \cup \{(\text{TagID}_i, \text{DIRECTION})\}$ 
5:   else
6:      $\mathcal{P}^{(t)} \leftarrow \mathcal{P}^{(t)} \cup \{\text{TagID}_i\}$ 
7:   end if
8: end for
9: if  $\mathcal{P}^{(t)} = \emptyset$  then
10:  return  $\emptyset, \mathcal{D}^{(t)}$ 
11: end if
12:  $\mathcal{X}^{(t)} \leftarrow \{s_i^{(t)} : \text{TagID}_i \in \mathcal{P}^{(t)}\}$ 
13:  $\mu^{(t)} \leftarrow \text{MEDIAN}(\mathcal{X}^{(t)})$ 
14:  $\text{MAD}^{(t)} \leftarrow \max(\text{MEDIAN}(\{|x - \mu^{(t)}| : x \in \mathcal{X}^{(t)}\}), \varepsilon)$ 
15:  $\rho^{(t)} \leftarrow \frac{|\mathcal{D}^{(t)}| + |\{i \in \mathcal{P}^{(t)} : |s_i^{(t)} - \mu^{(t)}| > \gamma_{\text{mad}} \text{MAD}^{(t)}\}|}{|\mathcal{S}^{(t)}|}$ 
16:  $\tau_{\min}^{(t)} \leftarrow \alpha(\rho^{(t)}) \mu^{(t)}$ 
17:  $\tau_{\max}^{(t)} \leftarrow \mu^{(t)} + \gamma_{\text{mad}} \text{MAD}^{(t)}$ 
18:  $\mathcal{A}^{(t)} \leftarrow \emptyset$ 
19: for all  $\text{TagID}_i \in \mathcal{P}^{(t)}$  do
20:   if  $\tau_{\min}^{(t)} \leq s_i^{(t)} \leq \tau_{\max}^{(t)}$  then
21:      $\mathcal{A}^{(t)} \leftarrow \mathcal{A}^{(t)} \cup \{\mathcal{U}^{(t)}[\text{TagID}_i]\}$ 
22:   else
23:      $\mathcal{D}^{(t)} \leftarrow \mathcal{D}^{(t)} \cup \{(\text{TagID}_i, \text{MAGNITUDE})\}$ 
24:   end if
25: end for
26: return  $\mathcal{A}^{(t)}, \mathcal{D}^{(t)}$ 

```

5.3.4 Phase IV: Secure Aggregation

The Aggregation Server adds only accepted ciphertext vectors and asks the TKA to decrypt the aggregate. Individual accepted gradients are not decrypted;

rejected submissions are discarded while still encrypted.

5.3.5 Phase V: Reference Update and Model Distribution

The final phase averages the accepted aggregate, updates the model, and refreshes the reference through momentum β . Normalisation keeps later FIP scores directional. If no client is accepted, the model and reference are left unchanged. Algorithm 5 combines these steps.

Secure Aggregation and Momentum-Based Reference Update

A geometric view of the final round update — accepted encrypted coordinates are homomorphically aggregated, only aggregate coordinates are decrypted by the TKA, the averaged global gradient updates the model, and a momentum-normalized trusted reference $R^{(t+1)}$ is stored for the next round.

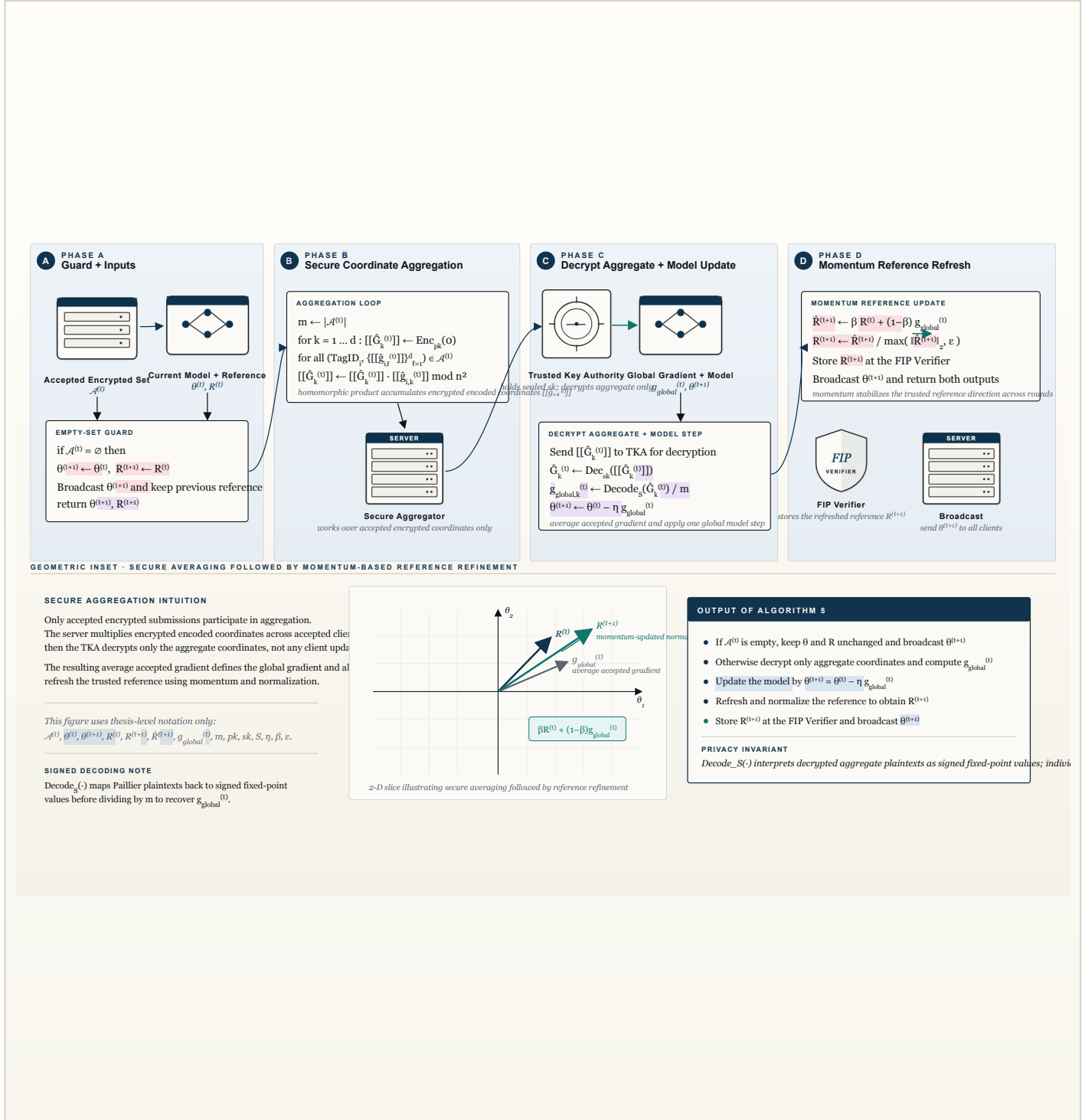


Figure 5. Algorithm 5. If the accepted encrypted set $\mathcal{A}^{(t)}$ is empty, the server keeps $\theta^{(t+1)} \leftarrow \theta^{(t)}$ and $R^{(t+1)} \leftarrow R^{(t)}$, broadcasts the unchanged model, and returns. Otherwise, for each coordinate k , the server homomorphically aggregates the accepted encrypted encoded values into $[[\hat{G}_k^{(t)}]]$ and sends only these aggregate ciphertexts to the TKA. After decryption and signed fixed-point decoding, the average accepted gradient $g_{\text{global}}^{(t)}$ is recovered and used to update the model by $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta g_{\text{global}}^{(t)}$. The trusted reference is then refreshed as $\hat{R}^{(t+1)} \leftarrow \beta R^{(t)} + (1-\beta)g_{\text{global}}^{(t)}$ and normalized to $R^{(t+1)}$ using $\max(\|\hat{R}^{(t+1)}\|_2, \epsilon)$. The inset illustrates secure averaging followed by momentum-based reference refinement.

Algorithm 5 Secure Aggregation and Momentum-Based Reference Update

Require: Accepted encrypted set $\mathcal{A}^{(t)}$, model $\theta^{(t)}$, reference $\mathbf{R}^{(t)}$, public key pk , secret key sk held by TKA, scale s , learning rate η , momentum β , stability floor ε

Ensure: Updated model $\theta^{(t+1)}$, updated reference $\mathbf{R}^{(t+1)}$

```

1: if  $\mathcal{A}^{(t)} = \emptyset$  then
2:    $\theta^{(t+1)} \leftarrow \theta^{(t)}$ ,  $\mathbf{R}^{(t+1)} \leftarrow \mathbf{R}^{(t)}$ 
3:   Broadcast  $\theta^{(t+1)}$  and keep the previous reference
4:   return  $\theta^{(t+1)}$ ,  $\mathbf{R}^{(t+1)}$ 
5: end if
6:  $m \leftarrow |\mathcal{A}^{(t)}|$ 
7: for  $k \leftarrow 1$  to  $d$  do
8:    $\llbracket \hat{G}_k^{(t)} \rrbracket \leftarrow \text{ENC}_{pk}(0)$ 
9:   for all  $(\text{TagID}_i, \{g_{i,\ell}^{(t)}\}_{\ell=1}^d) \in \mathcal{A}^{(t)}$  do
10:     $\llbracket \hat{G}_k^{(t)} \rrbracket \leftarrow \llbracket \hat{G}_k^{(t)} \rrbracket \cdot \llbracket g_{i,k}^{(t)} \rrbracket \bmod n^2$ 
11:   end for
12:   Send  $\llbracket \hat{G}_k^{(t)} \rrbracket$  to TKA for aggregate decryption
13:    $\hat{G}_k^{(t)} \leftarrow \text{DEC}_{sk}(\llbracket \hat{G}_k^{(t)} \rrbracket)$ 
14:    $g_{\text{global},k}^{(t)} \leftarrow \text{DECODE}_s(\hat{G}_k^{(t)}) / m$ 
15: end for
16:  $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \mathbf{g}_{\text{global}}^{(t)}$ 
17:  $\tilde{\mathbf{R}}^{(t+1)} \leftarrow \beta \mathbf{R}^{(t)} + (1 - \beta) \mathbf{g}_{\text{global}}^{(t)}$ 
18:  $\mathbf{R}^{(t+1)} \leftarrow \tilde{\mathbf{R}}^{(t+1)} / \max(\|\tilde{\mathbf{R}}^{(t+1)}\|_2, \varepsilon)$ 
19: Store  $\mathbf{R}^{(t+1)}$  at the FIP Verifier and broadcast  $\theta^{(t+1)}$ 
20: return  $\theta^{(t+1)}$ ,  $\mathbf{R}^{(t+1)}$ 

```

5.4 Theoretical Analysis

The three guarantees below formalise the privacy, admissibility, and descent conditions introduced in Section 4.6.

5.4.1 Gradient Privacy

Theorem 5.1 (Gradient Privacy). *In every round t , the FIP Verifier reveals exactly one scalar per client, namely the projection $s_i^{(t)} = \langle \mathbf{g}_i^{(t)}, \mathbf{R}^{(t)} \rangle \in \mathbb{R}$. For a gradient of dimension $d \gg 1$, recovering $\mathbf{g}_i^{(t)}$ from $s_i^{(t)}$ is information-theoretically impossible: the linear system $\mathbf{R}^{(t)\top} \mathbf{g}_i^{(t)} = s_i^{(t)}$ is under-determined and admits a $(d - 1)$ -dimensional affine solution space. Equivalently, the mutual information satisfies $I(\mathbf{g}_i^{(t)}; s_i^{(t)}) \leq H(s_i^{(t)}) \ll H(\mathbf{g}_i^{(t)})$.*

Proof. The FIP Verifier's only operation on ciphertext $[\![\mathbf{g}_i^{(t)}]\!]$ is the homomorphic inner product of (4.8), which yields $[s_i^{(t)}]$. Given $s_i^{(t)}$ and $\mathbf{R}^{(t)}$, the preimage of $\mathbf{g} \mapsto \langle \mathbf{g}, \mathbf{R}^{(t)} \rangle$ is a $(d - 1)$ -dimensional affine hyperplane. Gradients differing by a vector orthogonal to $\mathbf{R}^{(t)}$ are indistinguishable from the scalar alone, and the data-processing inequality gives $I(\mathbf{g}_i^{(t)}; s_i^{(t)}) \leq H(s_i^{(t)}) \ll H(\mathbf{g}_i^{(t)})$. $\square$

5.4.2 Attack Resilience

Theorem 5.2 (Attack Resilience). *Any malicious update $\mathbf{g}_{adv}$ accepted by the FIP-FL filter in round t must satisfy simultaneously*

$$\langle \mathbf{g}_{adv}, \mathbf{R}^{(t)} \rangle > 0 \quad (\text{direction constraint}), \quad (5.1)$$

$$\tau_{\min}^{(t)} \leq \langle \mathbf{g}_{adv}, \mathbf{R}^{(t)} \rangle \leq \tau_{\max}^{(t)} \quad (\text{magnitude constraint}), \quad (5.2)$$

where $\tau_{\min}^{(t)}$ and $\tau_{\max}^{(t)}$ are computed from the round-wise positive-score distribution. The two constraints jointly rule out direction-reversal, scaling-to-zero, and destabilising large-norm attacks while keeping the audit independent of any external auditor.

Proof. The geometric audit first applies the non-positive direction check and then the $\tau_{\min}^{(t)}/\tau_{\max}^{(t)}$ magnitude window. An adversary failing either clause is rejected. A direct poisoning gradient $\mathbf{g}_{adv} \approx -\mathbf{g}_{\text{honest}}$ gives a non-positive score, a near-zero scaling attack falls below $\tau_{\min}^{(t)}$, and a destabilising large-norm attack exceeds $\tau_{\max}^{(t)} = \mu^{(t)} + 5 \cdot \text{MAD}^{(t)}$. $\square$

5.4.3 Convergence Guarantee

Theorem 5.3 (Descent under L -Smoothness). *Let $F(\mathbf{w})$ be L -smooth and let $\Delta_{\text{global}}^{(t)}$ be the mean of the accepted updates in round t . If the accepted aggregate is descent-aligned with the true gradient,*

$$\langle \nabla F(\mathbf{w}^{(t)}), \Delta_{\text{global}}^{(t)} \rangle > 0, \quad (5.3)$$

then any learning rate

$$0 < \eta < \frac{2\langle \nabla F(\mathbf{w}^{(t)}), \Delta_{\text{global}}^{(t)} \rangle}{L\|\Delta_{\text{global}}^{(t)}\|_2^2} \quad (5.4)$$

gives a strict one-round loss decrease:

$$F(\mathbf{w}^{(t+1)}) < F(\mathbf{w}^{(t)}). \quad (5.5)$$

Consequently, if this alignment holds in expectation over accepted rounds, the expected global loss decreases monotonically.

Proof. The L -smoothness descent lemma gives

$$F(\mathbf{w}^{(t)} - \eta \Delta_{\text{global}}^{(t)}) \leq F(\mathbf{w}^{(t)}) - \eta \langle \nabla F(\mathbf{w}^{(t)}), \Delta_{\text{global}}^{(t)} \rangle + \frac{L\eta^2}{2} \|\Delta_{\text{global}}^{(t)}\|_2^2. \quad (5.6)$$

Under the stated alignment condition, the right-hand side is strictly below $F(\mathbf{w}^{(t)})$ whenever η satisfies the stated bound. The FIP-FL filter supports this condition by admitting only updates with positive reference projection and bounded scalar magnitude; the theorem states the resulting descent requirement explicitly rather than assuming the reference is perfectly equal to the true gradient. $\square$

Together, these results set the experimental checklist: hide the full gradient, reject harmful or over-scaled projections, and keep the accepted aggregate descent-oriented under the stated smoothness condition.

6 Experimental Setup

This chapter records the FIP-FL experimental protocol: datasets, partitions, attacks, metrics, implementation stack, and hardware.

6.1 Datasets

The evaluation uses one image benchmark and one intrusion-detection benchmark, summarized in Table 6.1.

Table 6.1: Datasets used in the evaluation of FIP-FL.

Dataset	Modality	Classes	Features	Samples
MNIST	Image	10	784	60,000/10,000
KDDCup99	Tabular	5	41	$\approx 494\text{K}$ (10%)

6

MNIST. MNIST contains 60,000 training and 10,000 test grayscale images of size 28×28 , labelled over digits $\{0, 1, \dots, 9\}$. I flatten each image to 784 features and normalize pixels to $[0, 1]$. Since Yazdinejad et al. [1] also report MNIST results, it is the direct auditor-baseline comparison point.

KDDCup99. KDDCup99 provides the tabular intrusion-detection case. I use the standard 10% subset of the 4.9 million-record corpus, approximately 494,000 records. Each record has 41 traffic and protocol features; labels are grouped into NORMAL, DoS, PROBE, R2L, and U2R. Categorical fields are one-hot encoded and continuous fields are standardized. Its heavy imbalance tests whether the score window handles uneven class priors.

Model architecture. Both datasets use a softmax logistic-regression classifier, matching the SenSys 2026 poster and the auditor-based baseline. The parameter dimension is $d = (\text{input_dim}) \cdot C + C$, including bias terms. This keeps Paillier encryption tractable with a 1024-bit modulus and matches the convex L -smooth setting of Theorem 5.3.

6.2 Data Partitioning

Each dataset is split into $N = 10$ disjoint client shards. In the IID regime, the shuffled training set is assigned to equal shards, so each client receives approximately $|\mathcal{D}_{\text{train}}|/N$ samples with the global label mix in expectation.

The non-IID regime uses label skew. For a C -class task, client i gets a primary label set $\mathcal{L}_i$ of size $\lceil C/2 \rceil$, with $\lfloor C/4 \rfloor$ overlap between consecutive clients. I assign 80% of each client's samples from $\mathcal{L}_i$ and draw the remaining 20% from all classes. This follows the SenSys 2026 protocol and Yazdinejad et al. [1]; it is difficult because honest gradients can differ under skew while still helping the global objective.

6.3 Attack Models

The evaluation uses two pre-training label-flipping attacks. Both modify local labels before gradient computation and do not require the Paillier key or FIP score.

Untargeted label flipping. A malicious client relabels each local sample by a fixed cyclic offset:

$$y_k^{(i, \text{mal})} = (y_k^{(i)} + \kappa) \bmod C. \quad (6.1)$$

For MNIST I use $\kappa = 5$, a no-fixed-point digit permutation. For KDDCup99 I use $\kappa = 2$ because the collapsed task has five classes. The attack makes malicious clients train toward a wrong classification objective.

Targeted label flipping. A malicious client changes only a source class c_s to a

target class c_t :

$$y_k^{(i, \text{mal})} = \begin{cases} c_t & \text{if } y_k^{(i)} = c_s, \\ y_k^{(i)} & \text{otherwise.} \end{cases} \quad (6.2)$$

For MNIST I use $(c_s, c_t) = (0, 7)$. For KDDCup99 I use the analogous source–target pair from the dominant skew-regime classes. Since the attack harms one class while leaving the rest plausible, target-class accuracy is the main diagnostic.

Malicious-client rates. For each dataset, partition, and attack type, I sweep $f/N \in \{10\%, 20\%, 30\%, 50\%\}$. The 50% row is the headline stress case near the limit where majority-based defences become fragile.

6.4 Evaluation Metrics

The results use five metrics: three measure held-out utility, and two measure verifier-side detection quality.

Target accuracy (T_{acc}). Accuracy restricted to test samples from the targeted source class c_s :

$$T_{\text{acc}} = \frac{|\{(x, y) \in \mathcal{D}_{\text{test}} : y = c_s, \hat{f}(x) = c_s\}|}{|\{(x, y) \in \mathcal{D}_{\text{test}} : y = c_s\}|}. \quad (6.3)$$

A successful targeted attack drives this value down; a successful defence keeps it high. For untargeted attacks, class 0 is used as the fixed diagnostic reference class.

Other-label accuracy (O_{acc}). Accuracy on all test samples outside the targeted source class:

$$O_{\text{acc}} = \frac{|\{(x, y) \in \mathcal{D}_{\text{test}} : y \neq c_s, \hat{f}(x) = y\}|}{|\{(x, y) \in \mathcal{D}_{\text{test}} : y \neq c_s\}|}. \quad (6.4)$$

This separates targeted damage from collateral loss on the rest of the task.

Overall accuracy (ACC). Standard test accuracy over all classes:

$$\text{ACC} = \frac{|\{(x, y) \in \mathcal{D}_{\text{test}} : \hat{f}(x) = y\}|}{|\mathcal{D}_{\text{test}}|}. \quad (6.5)$$

This is the primary utility metric for untargeted attacks.

True positive rate (TPR). Fraction of malicious clients rejected by the FIP-FL audit:

$$\text{TPR} = \frac{|\{i : \text{client } i \text{ is malicious and is rejected}\}|}{|\{i : \text{client } i \text{ is malicious}\}|}. \quad (6.6)$$

TPR measures malicious-client rejection; $\text{TPR} = 1.00$ means every malicious client in that run is filtered.

False positive rate (FPR). Fraction of honest clients rejected by the FIP-FL audit:

$$\text{FPR} = \frac{|\{i : \text{client } i \text{ is honest and is rejected}\}|}{|\{i : \text{client } i \text{ is honest}\}|}. \quad (6.7)$$

FPR measures the honest-client cost of the audit. All metrics are averaged over three independent seeds, and a standard deviation is reported when it exceeds 0.5 percentage points.

6.5 Implementation and Hardware

Experiments are implemented in Python 3 using NumPy, scikit-learn, and phe with a 1024-bit Paillier modulus. The FIP Verifier and Aggregation Server run as separate Python processes connected by in-memory queues; the N clients are worker processes. Encryption and FIP-score computation use multiprocessing.

Unless stated otherwise, all runs use $R = 300$, $B = 50$, $E = 1$, $\eta = 0.01$, fixed-point scale $s = 10,000$, direction threshold $\epsilon_{\text{dir}} = 0$, MAD multiplier $\gamma_{\text{mad}} = 5$, stability floor $\varepsilon = 10^{-6}$, and reference momentum $\beta = 0.80$. The adaptive lower-floor ratio $\alpha(\rho)$ is 0.55, 0.50, 0.40, and 0.30 for the four increasing attack-rate bands.

Wall-clock measurements in Chapter 8 use a workstation with a 25-core CPU, 128 GB RAM, and a Linux-based operating system. The $13\times$ speed-up cited in the abstract is measured on the Phase II homomorphic inner-product step against the naive single-core implementation.

7 Results and Analysis

This chapter reports the completed FIP-FL grid from Chapter 6. Tables use round-300 overall accuracy (ACC), target-class accuracy (T_{acc}), other-label accuracy (O_{acc}), true positive rate (TPR), and false positive rate (FPR). Accuracies are percentages; TPR and FPR are proportions.

7.1 Non-IID Partition, Untargeted Attacks

This is the hardest untargeted setting: label skew spreads honest gradients, and malicious clients add gradients trained on cyclically flipped labels. Overall accuracy is the utility metric; TPR and FPR show filtering cost.

7.1.1 MNIST

Table 7.1: MNIST, non-IID partition, untargeted label-flipping attack.

AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	91.22	97.60	90.18	1.00	0.00
20%	91.02	97.87	89.94	1.00	0.00
30%	90.78	97.33	89.31	1.00	0.00
50%	91.10	97.51	89.91	1.00	0.20

7.1.2 KDDCup99

Table 7.2: KDDCup99, non-IID partition (Dirichlet $\alpha = 0.5$), untargeted label-flipping attack.

AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	99.71	99.50	84.62	1.00	0.00
20%	99.72	99.55	83.65	1.00	0.00
30%	99.69	99.45	85.58	1.00	0.14
50%	99.69	99.70	77.88	1.00	0.40

Observations. In both non-IID untargeted tables, TPR is 1.00 for every attack rate. MNIST accuracy stays within 90.78–91.22%, with false rejection only at 50% attack rate. KDDCup99 remains above 99.6% accuracy; higher FPR at the heaviest load reflects the difficulty of separating honest label-skewed gradients from weak positive malicious scores.

7.2 Non-IID Partition, Targeted Attacks

Targeted label flipping tries to damage the source class while keeping the rest of the classifier useful, so the primary diagnostic is T_{acc} .

7.2.1 MNIST

Table 7.3: MNIST, non-IID, targeted label-flipping attack ($c_s = 0, c_t = 7$); SenSys 2026 headline targeted-accuracy cell.

AR	Undefended (%)	Auditor-based (%)	FIP-FL (%)
50%	≈ 7.00	96.50	97.69

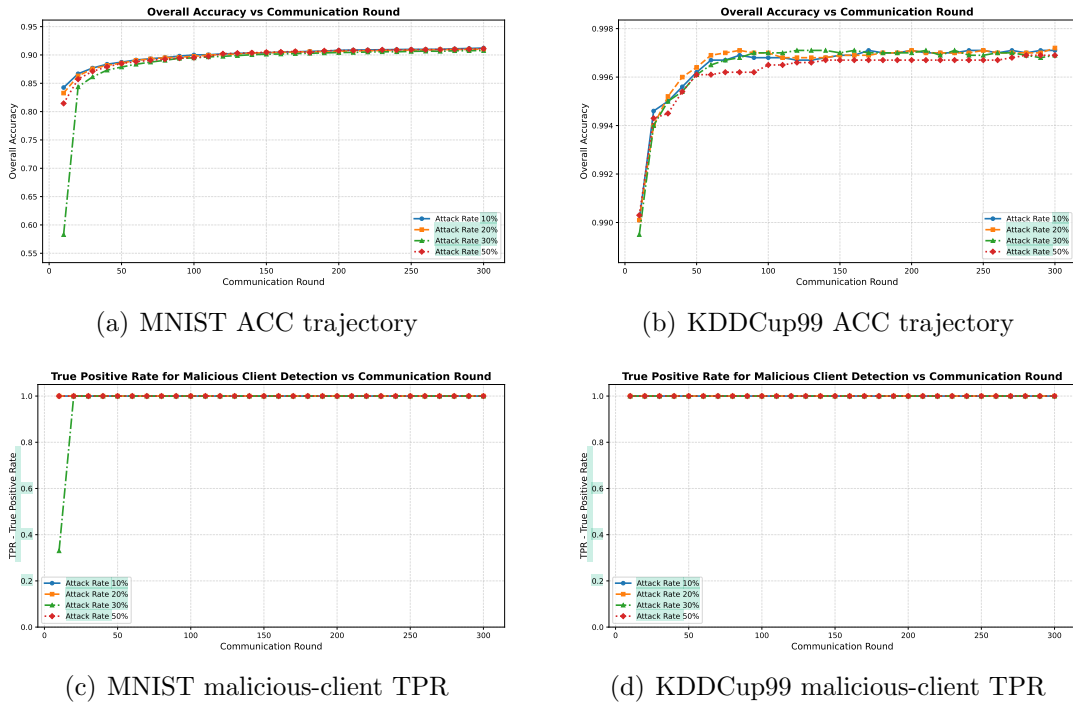


Figure 7.1: Non-IID untargeted attack dynamics over 300 communication rounds. Accuracy converges stably for both datasets, while the direction/magnitude audit reaches perfect malicious-client detection by the final round.

7.2.2 KDDCup99

Table 7.4: KDDCup99, non-IID partition (2 shards/client), targeted label-flipping attack.

AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	99.54	99.75	63.46	1.00	0.00
20%	99.69	99.65	78.85	1.00	0.00
30%	99.70	99.50	82.69	1.00	0.00
50%	99.52	98.41	86.54	1.00	0.00

Observations. The MNIST row is the accepted-poster headline result: at 50% malicious clients, FIP-FL raises T_{acc} from roughly 7% undefended to 97.69%, above the 96.5% auditor-based value of Yazdinejad et al. [1]. On KDDCup99, TPR remains 1.00, FPR remains 0.00, and $T_{\text{acc}} = 98.41\%$ even at 50% malicious

clients.

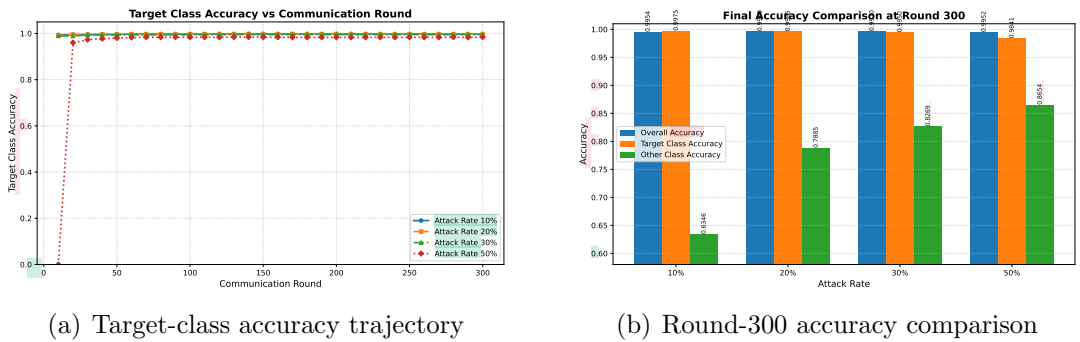


Figure 7.2: KDDCup99 non-IID targeted attack results. The target-class trajectory remains high through training, and the round-300 comparison shows that overall and target-class accuracy stay stable even as the malicious-client fraction increases.

7.3 IID Partition, Untargeted Attacks

IID partitions are the sanity-check regime: honest clients are statistically similar, so the FIP score distribution should be tighter than in label-skewed runs.

7.3.1 MNIST

Table 7.5: MNIST, IID partition, untargeted label-flipping attack.

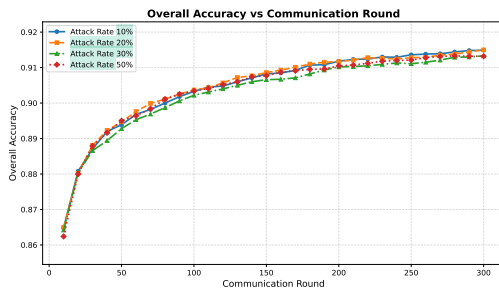
AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	91.49	97.51	90.40	1.00	0.00
20%	91.50	97.51	90.39	1.00	0.00
30%	91.33	97.42	90.28	1.00	0.00
50%	91.32	97.42	90.22	1.00	0.00

7.3.2 KDDCup99

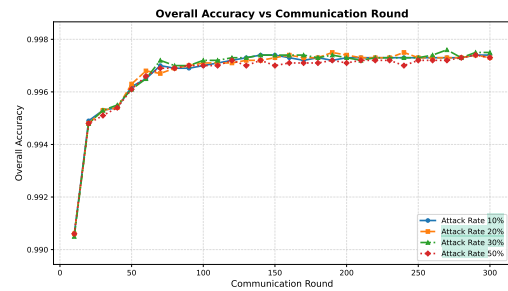
Table 7.6: KDDCup99, IID partition, untargeted label-flipping attack.

AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	99.74	99.60	85.58	1.00	0.00
20%	99.73	99.55	85.58	1.00	0.00
30%	99.75	99.60	85.58	1.00	0.00
50%	99.73	99.55	85.58	1.00	0.00

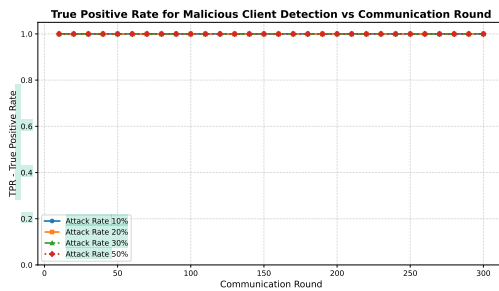
Observations. The IID untargeted regime behaves as expected: TPR is 1.00 and FPR is 0.00 for both datasets. MNIST ACC varies by less than 0.2 percentage points, and KDDCup99 metrics are almost unchanged across the sweep.



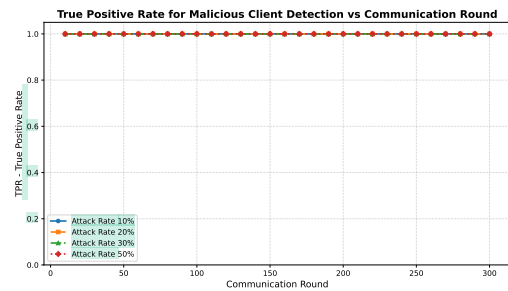
(a) MNIST ACC trajectory



(b) KDDCup99 ACC trajectory



(c) MNIST malicious-client TPR



(d) KDDCup99 malicious-client TPR

Figure 7.3: IID untargeted attack dynamics over 300 communication rounds. Both datasets show stable convergence, and the audit maintains perfect malicious-client detection without false positives at the reported final round.

7.4 IID Partition, Targeted Attacks

7.4.1 MNIST

Table 7.7: MNIST, IID partition, targeted label-flipping attack ($c_s = 0, c_t = 7$).

AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	91.62	97.51	90.57	1.00	0.00
20%	91.42	97.42	90.35	1.00	0.00
30%	91.32	97.42	90.22	1.00	0.00
50%	91.43	97.42	90.35	1.00	0.00

7.4.2 KDDCup99

Table 7.8: KDDCup99, IID partition, targeted label-flipping attack.

AR	ACC (%)	T_{acc} (%)	O_{acc} (%)	TPR	FPR
10%	99.73	99.60	84.62	1.00	0.00
20%	99.73	99.55	85.58	1.00	0.00
30%	99.72	99.50	85.58	1.00	0.00
50%	99.73	99.55	85.58	1.00	0.00

Observations. IID targeted runs match the IID untargeted pattern: exact detection, no final-round false rejection, MNIST $T_{\text{acc}} \geq 97.42\%$, and KDDCup99 ACC in the 99.72–99.73% range with $T_{\text{acc}} \geq 99.50\%$.

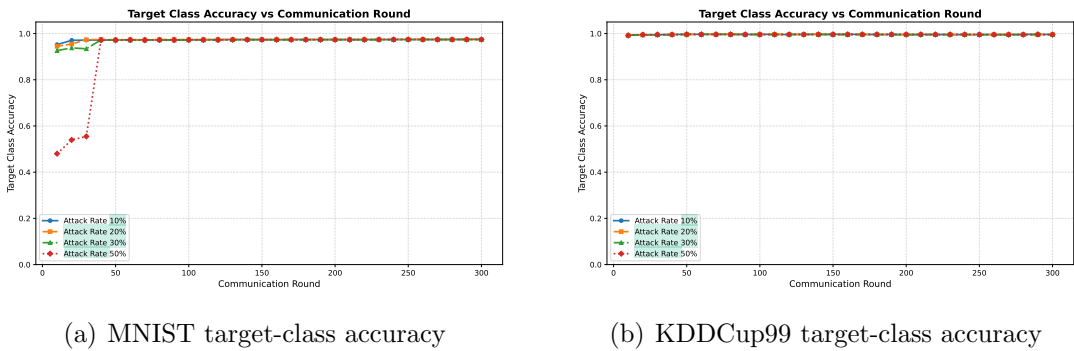


Figure 7.4: IID targeted attack target-class accuracy over 300 communication rounds. In both datasets, the target class recovers to a stable high-accuracy regime by the final round.

7.5 Aggregate Findings Across the Grid

The completed grid supports three conclusions. First, every reported row reaches $\text{TPR} = 1.00$, including the 50% malicious-client cases. Second, false rejection appears only under heavy non-IID untargeted load, and accuracy remains stable there. Third, targeted and untargeted attacks are both captured by the same scalar FIP audit in the evaluated settings. Chapter 8 compares these results with the auditor-based baseline.

8 Comparative Analysis

This chapter compares FIP-FL with the defences from Chapter 3, especially the auditor-based baseline of Yazdinejad et al. [1]. The comparison focuses on decrypted information, verifier work, post-filter communication, non-IID robustness, and MNIST non-IID targeted accuracy.

8.1 Privacy

The auditor-based scheme decrypts cluster-level statistics in each round: the projected Gaussian-mixture representation and the distance statistics used for screening. Even without reconstructing individual gradients, cluster means and covariances can reveal population structure, especially under non-IID data.

FIP-FL decrypts one scalar per client per round, $s_i^{(t)} = \langle \mathbf{g}_i^{(t)}, \mathbf{R}^{(t)} \rangle$. It does not decrypt cluster means, covariance matrices, projection coefficients, or pairwise distances; the model update is obtained only after accepted ciphertexts are aggregated.

8.2 Computational Complexity

Table 8.1: Per-round computational cost. N : clients; $|\mathcal{A}|$: accepted clients; d : model dimension; B : batch size; E : local epochs.

Entity	Auditor-based [1]	FIP-FL
Client	$\mathcal{O}(dBE)$	$\mathcal{O}(dBE)$
Verifier / Auditor	$\mathcal{O}(NKD + Nd \log d) + \mathcal{O}(d^3)$	$\mathcal{O}(Nd + N \log N)$
Aggregation Server	$\mathcal{O}(dN \log N)$	$\mathcal{O}(\mathcal{A} d)$

Client-side cost is unchanged because both schemes run local SGD. The auditor-based method pays for projection, mixture fitting, and covariance inversion, while FIP-FL computes N homomorphic inner products and median/MAD statistics. After filtering, the server sums only accepted ciphertext vectors, giving $\mathcal{O}(|\mathcal{A}|d)$ work.

8.3 Communication Complexity

Table 8.2: Per-round communication cost by directed link.

Link	Auditor-based [1]	FIP-FL
Client $\rightarrow$ Verifier	$\mathcal{O}(Nd)$	$\mathcal{O}(Nd)$
Verifier $\rightarrow$ Server	$\mathcal{O}(L(Z + N))$	$\mathcal{O}(\mathcal{A} d)$
Server $\rightarrow$ Clients	$\mathcal{O}(d)$	$\mathcal{O}(d)$

The initial client-to-verifier payload is the same because both designs submit encrypted updates. After verification, FIP-FL forwards only accepted ciphertexts and avoids the extra auditor path.

8.4 Robustness and Engineering Trade-Offs

The auditor-based detector works best when honest clients form a tighter and more populous cluster than malicious clients; label skew weakens that assumption. FIP-FL instead tests each update against a trusted reference and an adaptive scalar window.

The reference update is conservative: only updates passing both checks enter the aggregate, and the next reference uses exponential moving average rather than full replacement. Phase II FIP computations are independent across clients, giving the $13\times$ wall-clock speed-up on the 25-core workstation of Section 6.5.

8.5 Scheme-versus-Scheme Accuracy Comparison

Table 8.3: Accuracy on MNIST non-IID under a 50% malicious-client rate. Entries are percentages; baseline values are from [1].

Scheme	Targeted (T_{acc})	Untargeted (ACC)
Krum [8]	71.4	78.3
Auror [11]	32.5	49.9
PEFL [13]	46.4	57.4
Sybils / FoolsGold [15]	89.7	89.5
FL-Trust [12]	37.8	43.4
ShieldFL [16]	90.1	89.4
Auditor-based [1]	96.5	96.3
FIP-FL (this thesis)	97.69	88.82

FIP-FL gives the best targeted accuracy in the table while using weaker trust assumptions than the auditor-based baseline. Its untargeted result is close to

FoolsGold and ShieldFL but below the auditor-based row, making untargeted recovery the clearest empirical target for future work.

9 Conclusion and Future Work

9.1 Summary of the Thesis

This thesis examined whether Byzantine filtering can be done inside PPFL without an auditor that decrypts cluster statistics. FIP-FL replaces cluster-level inspection with a functional inner product between each encrypted update and a momentum-tracked reference direction. The decrypted value is only a scalar score, checked first for direction and then against a robust magnitude window.

The framework combines Paillier encryption, homomorphic score computation, secure aggregation, and momentum-based reference updating. The analysis states what the scalar score reveals, what conditions an accepted adversary must satisfy, and when the accepted aggregate gives descent. In the completed grid, FIP-FL detects every reported malicious-client configuration, raises MNIST non-IID target accuracy from roughly 7% undefended to 97.69% at 50% malicious clients, removes the auditor's $\mathcal{O}(d^3)$ covariance step, and gives a 13 $\times$ Phase II speed-up.

9.2 Limitations and Future Work

FIP-FL deliberately uses one reference direction per round. This keeps verification linear, but highly heterogeneous federations may contain several honest descent directions. A multi-reference cone $\{\mathbf{R}_k^{(t)}\}_{k=1}^K$ is a natural extension, provided it includes safeguards against cone-overlap exploits.

The present evaluation uses logistic-regression models, dimensions in the low thousands, a fixed client set, and one attack mode per run. Deep-network deployment would likely require packed homomorphic encryption such as BFV or CKKS [5]. A broader study should include client churn, changing data

distributions, adaptive adversaries, and mixed attacks.

Finally, the initial reference is computed from a small public bootstrap set. Future work could derive the first reference from robust aggregate statistics while still avoiding individual-gradient decryption. End-to-end deployment would then test the balance among encryption cost, communication cost, and model utility.

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